

Canonical Form of Antisymmetric Matrices

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(Dated: 06/24/2003)

Abstract

The canonical geminal coefficients often advertised as "canonical" for a given geminal are in fact very non unique. In this paper we show by carefully examining a Theorem by B. Zumino how to construct unique canonical coefficients for a geminal. This result leads to expansions of Antisymmetrised Geminal Power States solely in terms of Natural Generalized Spin Orbitals and geminal occupation numbers.

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I. INTRODUCTION

Antisymmetric Geminal Power (AGP) states provide an unbiased way to treat correlation for both strongly and weakly correlated systems. This has been displayed by their use as a reference state for the polarization propagator $[[[?]]]$ in the simultaneous production of dissociation curves for several low lying excited states of light diatomic molecules which were in good agreement with experimental results. Although AGP states are rarely the exact solutions of the Schrödinger equation they can serve as an unbiased approximation ansatz generalizing the Independent Particle State (IPS) approach by including a better description of electron - electron interaction, while at the same time maintaining a balanced and physically interpretable model of both molecular and extended systems. In this paper we in fact show that an $2N$ -electron AGP state is completely determined by geminal occupation numbers and Natural General Spin Orbitals (NGSOs), two quantities that have a clear physical significance. A $2N$ -electron AGP state is the antisymmetrized product of

N identical 2-electron geminals thus the properties and form of the geminal determine the characteristics of the product state e.g. the expectation value expressions and will form the focus of this paper.

The methods used to energy optimize an AGP state in the previously cited references were not numerically robust nor easily extendable to large systems. Recently we have [?]] described a method that overcomes these numerical difficulties and can be applied to molecules of various sizes. A salient feature of this procedure is that the geminal parameters always refer to a *fixed* localized atomic orbital basis, which allows one to visualize the AGP state in a chemical intuitive manner and introduce approximations to the integral matrix in a natural way. At the same time the various expressions for the energy and its gradient are more conveniently formulated in terms of the canonical form (canonical orbitals and coefficients) of the geminal, which was first discussed in a quantum chemical context in the works of [?],[?].

Producing the canonical form of a geminal is equivalent to bringing an antisymmetric *complex* matrix to a canonical form. This however is not a straightforward task as such matrices are *not* normal [?]] (therefore one cannot assume they are diagonalizable) and the methods used for real antisymmetric matrices have to be carefully modified. However Zumino [?]] constructively proved a theorem showing how to accomplish this transformation. In nearly all discussions of the properties of a geminal various subtleties of the canonical form have been missed or not focused on, in particular there is *no unique* set of canonical coefficients nor canonical orbitals. The latter observation is not surprising and since the canonical orbitals have always known to be indeterminate up to at least 2×2 special unitary transformations, nevertheless the fact that there is not a unique set of canonical coefficients has not been widely advertised nor studied in depth. However a corollary in the Zumino paper shows how one can define a unique set of canonical coefficients as the positive square roots of the eigenvalues of the First Order Reduced Density Operator (FORDO) of the 2-particle geminal state, which in turn defines canonical orbitals up to special unitary transformations that mix NGSOs associated with a given occupation number.

After the introduction (section I) we define in section II the geminal and reduced density operator associated with this 2 -particle state and we also review some aspects of tensor product expansions that can lead to some ambiguity in the normalization of the geminal expansion coefficients, in section III we develop Zumino's theorem and two corollaries in

detail, while in section IV we describe a procedure for producing the physically significant canonical coefficients and orbitals and examine the subtleties of numerically implementing this procedure and we conclude with a discussion in section V.

II. GEMINALS AND ANTISYMMETRIC COMPLEX MATRICES

In this section we describe how one forms the matrix of Natural Generalized Spin Orbitals (NGSO) that brings the complex antisymmetric matrix of geminal coefficients $\mathbf{c}$ to block diagonal form (see section III). It is important to note the significance of the term "Generalized" which means that the orbitals have both alpha and beta spin components. This is necessary on two fronts, one mathematical in order to obtain the correct pairing associated with antisymmetry and one practical in order to form the most flexible approximation ansatz. In order to produce the NGSOs one needs to construct the FORDO of the 2-particle geminal state $|g\rangle$ and then sum the indices relating to one of the particles. This operation is straightforward in terms of a general tensor product basis (see Appendix A), but introduces some subtleties of normalization when one uses the standard expression of the geminal in terms of an antisymmetrized tensor product basis. For the purposes of clarity we compare the use of these two expansions.

A normalized geminal $|g\rangle$ can be expanded in terms of a general tensor product orthonormal basis $\{|\varphi_{j_1} \otimes \varphi_{j_2}\rangle; 1 \leq j_1, j_2 \leq r\}$ as

$$|g\rangle = \frac{1}{2} \sum_{1 \leq j_1, j_2 \leq r} c_{j_1 j_2} |\varphi_{j_1} \otimes \varphi_{j_2}\rangle; \quad c_{j_1 j_2} = -c_{j_2 j_1}, \quad c_{j_1 j_2} \in \mathbb{C} \quad (1)$$

where $\{|\varphi_j\rangle; 1 \leq j \leq r\}$ is a complete orthonormal basis for the one particle Hilbert space $\mathcal{H}^1$, and $\mathbb{C}$ denotes the complex numbers. The associated density operator to the state $|g\rangle$ is

$$|g\rangle \langle g| = \frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} c_{j_1 j_2} c_{k_1 k_2}^* |\varphi_{j_1} \otimes \varphi_{j_2}\rangle \langle \varphi_{k_1} \otimes \varphi_{k_2}| \quad (2)$$

As $\langle g|g\rangle = 1$ one observes that

$$\text{Tr} \{\mathbf{c} \mathbf{c}^\dagger\} = 4 \quad (3)$$

The expansion in terms of the tensor product basis $\{|\varphi_{j_1} \otimes \varphi_{j_2}\rangle\}$ can be compared to

$$|g\rangle = \frac{1}{2} \sum_{1 \leq j_1, j_2 \leq r} \tilde{c}_{j_1 j_2} |\varphi_{j_1} \wedge \varphi_{j_2}\rangle; \quad \tilde{c}_{j_1 j_2} = -\tilde{c}_{j_2 j_1} \quad (4)$$

in terms of the antisymmetrized tensor product basis $\{|\varphi_{j_1} \wedge \varphi_{j_2}\rangle; 1 \leq j_1 < j_2 \leq r\}$ where

$$|\varphi_{j_1} \wedge \varphi_{j_2}\rangle = \frac{1}{\sqrt{2}} (|\varphi_{j_1} \otimes \varphi_{j_2}\rangle - |\varphi_{j_2} \otimes \varphi_{j_1}\rangle) \quad (5)$$

Expressing the antisymmetric expansion in terms of the general one we obtain

$$\begin{aligned} \frac{1}{2} \sum_{1 \leq j_1, j_2 \leq r} \tilde{c}_{j_1 j_2} |\varphi_{j_1} \wedge \varphi_{j_2}\rangle &= \frac{1}{2} \sum_{1 \leq j_1, j_2 \leq r} \tilde{c}_{j_1 j_2} \frac{1}{\sqrt{2}} (|\varphi_{j_1} \otimes \varphi_{j_2}\rangle - |\varphi_{j_2} \otimes \varphi_{j_1}\rangle) \\ &= \frac{1}{2} \frac{1}{\sqrt{2}} \sum_{1 \leq j_1, j_2 \leq r} \tilde{c}_{j_1 j_2} |\varphi_{j_1} \otimes \varphi_{j_2}\rangle - \frac{1}{2} \frac{1}{\sqrt{2}} \sum_{1 \leq j_1, j_2 \leq r} \tilde{c}_{j_1 j_2} |\varphi_{j_2} \otimes \varphi_{j_1}\rangle \\ &= \frac{1}{2} \frac{1}{\sqrt{2}} \sum_{1 \leq j_1, j_2 \leq r} \tilde{c}_{j_1 j_2} |\varphi_{j_1} \otimes \varphi_{j_2}\rangle + \frac{1}{2} \frac{1}{\sqrt{2}} \sum_{1 \leq j_1, j_2 \leq r} \tilde{c}_{j_1 j_2} |\varphi_{j_1} \otimes \varphi_{j_2}\rangle \\ &= \frac{2}{\sqrt{2}} \sum_{1 \leq j_1, j_2 \leq r} \tilde{c}_{j_1 j_2} |\varphi_{j_1} \otimes \varphi_{j_2}\rangle \end{aligned} \quad (6)$$

which shows that

$$c_{j_1 j_2} = \sqrt{2} \tilde{c}_{j_1 j_2} \quad (7)$$

The contraction map $L_2^1[?] [?] [?]$ acting on $|g\rangle \langle g|$ produces

$$\begin{aligned} L_2^1(|g\rangle \langle g|) &= \frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} c_{j_1 j_2} c_{k_1 k_2}^* \langle \varphi_{k_2} | \varphi_{j_2} \rangle |\varphi_{j_1}\rangle \langle \varphi_{k_1}| = \frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1, k_2 \leq r}} c_{j_1 j_2} c_{k_1 k_2}^* \delta_{k_2 j_2} |\varphi_{j_1}\rangle \langle \varphi_{k_1}| \\ &= \frac{1}{4} \sum_{\substack{1 \leq j_1, j_2 \leq r \\ 1 \leq k_1 \leq r}} c_{j_1 j_2} c_{k_1 j_2}^* |\varphi_{j_1}\rangle \langle \varphi_{k_1}| = \frac{1}{4} \sum_{1 \leq j_1, j_2 \leq r} (\mathbf{c} \mathbf{c}^\dagger)_{j_1 k_1} |\varphi_{j_1}\rangle \langle \varphi_{k_1}| \end{aligned} \quad (8)$$

The First Order Reduced Density Operator (FORDO) $D^1(g)$ normalized to the number of electrons in the system, (in this case 2), is then given by

$$D^1(g) = 2L_2^1(|g\rangle \langle g|) \quad (9)$$

and the First Order Reduced Density Matrix (FORDM), $\mathbb{D}_\varphi^1(g)$, which is the *representation* of $D^1(g)$ in the $\{\varphi\}$ basis by

$$\mathbb{D}_\varphi^1(g) = \frac{1}{2} \mathbf{c} \mathbf{c}^\dagger = \tilde{\mathbf{c}} \tilde{\mathbf{c}}^\dagger \quad (10)$$

both of which have trace equal to 2 i.e.

$$\text{Tr} \{D^1(g)\} = \frac{1}{2} \text{Tr} \{\mathbf{c} \mathbf{c}^\dagger\} = \text{Tr} \{\tilde{\mathbf{c}} \tilde{\mathbf{c}}^\dagger\} = 2 \quad (11)$$

One can also observe that

$$\mathbb{D}_\varphi^1(g)^* = \frac{1}{2} \mathbf{c}^\dagger \mathbf{c} \quad (12)$$

and as $\mathbf{c}$ is antisymmetric

$$\mathbb{D}_\varphi^1(g) = \frac{1}{2}\mathbf{c}\mathbf{c}^\dagger = -\frac{1}{2}\mathbf{c}\mathbf{c}^* = \frac{1}{2}\mathbf{c}^t\mathbf{c}^* \quad (13)$$

The NGSO's $\{|y_j\rangle; 1 \leq j \leq r\}$ of $D^1(g)$ have expansion coefficients $\{\mathbf{y}_j; 1 \leq j \leq r\}$ in the $\{\varphi\}$ basis that are the eigenvectors of $\mathbb{D}_\varphi^1(g)$ given by

$$\mathbb{D}_\varphi^1(g) \mathbf{y}_j = n_j \mathbf{y}_j \quad (14)$$

leading to

$$\mathbb{Y}^\dagger \mathbb{D}_\varphi^1(g) \mathbb{Y} = \begin{pmatrix} \mathbf{n} & \mathbf{0} \\ \mathbf{0} & \mathbf{n} \end{pmatrix} \quad (15)$$

where

$$\mathbf{n} = \begin{pmatrix} n_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & n_s \end{pmatrix}; \quad n_j \geq 0, \quad r = 2s \quad (16)$$

and

$$\mathbb{Y}^\dagger \mathbb{Y} = \mathbb{Y} \mathbb{Y}^\dagger = \mathbb{I}_r \quad (17)$$

Although this transformation brings $\mathbb{D}_\varphi^1(g)$ to diagonal form it does *not* in general bring $\mathbf{c}$ to a canonical form. In the next section we show, how using a Theorem due to Zumino [?] one can achieve this result.

III. ZUMINOS' THEOREM

In this section we first describe Zuminos' Theorem and its proof in detail and then discuss an important corollary that leads to the definition of a unique, physically meaningful, canonical form of $\mathbf{c}$.

Theorem 1 *If $\mathbf{c}$ is a $2s \times 2s$ complex antisymmetric matrix then there exists a unitary matrix $\mathbb{T}$ such that*

$$\mathbb{T}\mathbf{c}\mathbb{T}^t = \begin{pmatrix} \mathbf{0} & \zeta \\ -\zeta & \mathbf{0} \end{pmatrix} \quad (18)$$

where

$$\zeta = \begin{pmatrix} \zeta_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \zeta_s \end{pmatrix}; \quad \zeta_j \in \mathbb{C}, \quad 1 \leq j \leq s \quad (19)$$

Proof. We first note some relationships implied by the antisymmetry of $\mathbf{c}$

$$\mathbf{c}^\dagger \mathbf{c} = -\mathbf{c}^* \mathbf{c} = (\mathbf{c} \mathbf{c}^\dagger)^t = (\mathbf{c} \mathbf{c}^\dagger)^* \quad (20)$$

(so $\mathbf{c}$ is not a normal operator in general) and one by the associativity of matrix product

$$\mathbf{c} (\mathbf{c}^\dagger \mathbf{c}) = (\mathbf{c} \mathbf{c}^\dagger) \mathbf{c} \quad (21)$$

which by using eq. (20) gives

$$(\mathbf{c} \mathbf{c}^\dagger) \mathbf{c} = \mathbf{c} (\mathbf{c} \mathbf{c}^\dagger)^* \quad (22)$$

Then we introduce $\mathbf{c}_U \mathbf{c}_U^\dagger$ through the definition

$$\mathbf{c} = \mathbb{U} \mathbf{c}_U \mathbb{U}^t \quad (23)$$

$$\mathbf{c}_U = \mathbb{U}^\dagger \mathbf{c} \mathbb{U}^* \quad (24)$$

where $\mathbb{U}$ is any unitary matrix and obtain that

$$\begin{aligned} \mathbf{c} \mathbf{c}^\dagger &= \mathbb{U} \mathbf{c}_U \mathbb{U}^t \mathbb{U}^* \mathbf{c}_U^\dagger \mathbb{U}^\dagger = \mathbb{U} \mathbf{c}_U \mathbf{c}_U^\dagger \mathbb{U}^\dagger \\ \mathbf{c}_U \mathbf{c}_U^\dagger &= \mathbb{U}^\dagger \mathbf{c} \mathbf{c}^\dagger \mathbb{U} \end{aligned} \quad (25)$$

and as $\mathbf{c}_U$ is also antisymmetric

$$\mathbf{c}_U \mathbf{c}_U^\dagger = -\mathbf{c}_U \mathbf{c}_U^* = (\mathbf{c}_U^\dagger \mathbf{c}_U)^t = (\mathbf{c}_U^\dagger \mathbf{c}_U)^*. \quad (26)$$

Hence we have that

$$(\mathbf{c} \mathbf{c}^\dagger) \mathbf{c} - \mathbf{c} (\mathbf{c} \mathbf{c}^\dagger)^* = (\mathbf{c}_U \mathbf{c}_U^\dagger) \mathbf{c}_U - \mathbf{c}_U (\mathbf{c}_U \mathbf{c}_U^\dagger)^* = 0 \quad (27)$$

We can choose $\mathbb{U}$ to diagonalize the hermitian matrix $\mathbf{c} \mathbf{c}^\dagger$ i.e.

$$\mathbb{U}^\dagger \mathbf{c} \mathbf{c}^\dagger \mathbb{U} = \begin{pmatrix} n_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & n_r \end{pmatrix} = \mathbf{c}_U \mathbf{c}_U^\dagger; \quad n_r \geq n_{r-1} \geq \dots \geq n_1 \geq 0. \quad (28)$$

Now using the relationships in eq. (27) for this $\mathbf{c}_U$ we obtain

$$\sum_{1 \leq j \leq r} \{n_i \delta_{ij} c_{Ujk} - c_{Uij} n_j \delta_{jk}\} = 0 \Rightarrow (n_i - n_k) c_{Uik} = 0; \quad 1 \leq i, k \leq r \quad (29)$$

which shows that

$$n_k \neq n_i \Rightarrow c_{Uik} = 0; \quad 1 \leq i \neq k \leq r \quad (30)$$

The diagonal elements

$$c_{Uii} = 0; \quad 1 \leq i \leq r \quad (31)$$

are always zero as $\mathbf{c}_U$ is antisymmetric. Thus the matrix $\mathbf{c}_U$ breaks into submatrices $\{\mathbb{C}_\kappa; 1 \leq \kappa \leq p\}$ corresponding to groups of n 's that are equal. Consider the partitioning of $\{1, \dots, r\}$ into p groups that correspond to equal n 's i.e. $\{\{1, \dots, m_1\}, \dots, \{m_{p-1} + 1, \dots, r\}\}$ where

$$\begin{aligned} n_1 &= n_2 = \dots = n_{m_1} = \eta_1 \\ &\vdots \\ n_{m_{p-1}+1} &= n_{m_{p-1}+2} = \dots = n_r = \eta_p \end{aligned} \quad (32)$$

then the matrix $\mathbf{c}_U$ has the block diagonal form

$$\mathbf{c}_U = \begin{pmatrix} \mathbf{c}_1 & \mathbb{O} & \dots & \mathbb{O} \\ \mathbb{O} & \ddots & \mathbb{O} & \vdots \\ \vdots & \mathbb{O} & \ddots & \mathbb{O} \\ \mathbb{O} & \dots & \mathbb{O} & \mathbf{c}_p \end{pmatrix} \quad (33)$$

and

$$\mathbf{c}_\kappa^\dagger \mathbf{c}_\kappa = \eta_\kappa \mathbb{I}_{d_\kappa}; \quad 1 \leq \kappa \leq p \quad (34)$$

where d_κ is the dimension of κ^{th} -block.

If $\eta_\kappa = 0$ then

$$\sum_{k \in \{m_{\kappa-1}+1, \dots, m_\kappa\}} c_{\kappa ik} c_{\kappa jk}^* = 0 \quad \forall i, j \in \{m_{\kappa-1} + 1, \dots, m_\kappa\} \quad (35)$$

so in particular

$$\sum_{k \in \{m_{\kappa-1}+1, \dots, m_\kappa\}} |c_{\kappa jk}|^2 = 0 \quad \forall j \in \{m_{\kappa-1} + 1, \dots, m_\kappa\} \quad (36)$$

which implies that $c_{\kappa jk} = 0 \quad \forall j, k \in \{m_{\kappa-1} + 1, \dots, m_\kappa\}$.

If $\eta_\kappa \neq 0$ then $\mathbb{B}_\kappa = \frac{1}{\sqrt{\eta_\kappa}} \mathbf{c}_\kappa$ is a $d_\kappa \times d_\kappa$ antisymmetric unitary matrix

$$\begin{aligned} \mathbb{B}_\kappa^\dagger \mathbb{B}_\kappa &= \mathbb{B}_\kappa \mathbb{B}_\kappa^\dagger = \mathbb{I}_{d_\kappa} \\ \Rightarrow \mathbb{B}_\kappa^* \mathbb{B}_\kappa &= \mathbb{B}_\kappa \mathbb{B}_\kappa^* \end{aligned} \quad (37)$$

i.e. $[\mathbb{B}_\kappa, \mathbb{B}_\kappa^\dagger] = [\mathbb{B}_\kappa, \mathbb{B}_\kappa^*] = 0$ showing that $\mathbb{B}_\kappa$ is a normal matrix. One can then define the matrices

$$\begin{aligned}\mathbb{N}_\kappa &= \frac{1}{2} \{\mathbb{B}_\kappa + \mathbb{B}_\kappa^*\} \\ \mathbb{M}_\kappa &= \frac{-i}{2} \{\mathbb{B}_\kappa - \mathbb{B}_\kappa^*\}\end{aligned}\quad (38)$$

which are not in general self adjoint but are real antisymmetric and commute with each other so that

$$\mathbb{B}_\kappa = \mathbb{N}_\kappa + i\mathbb{M}_\kappa. \quad (39)$$

These matrices can be simultaneously transformed into the canonical forms

$$\begin{pmatrix} \mathbf{0} & \alpha_\kappa \\ -\alpha_\kappa & \mathbf{0} \end{pmatrix} \text{ and } \begin{pmatrix} \mathbf{0} & \beta_\kappa \\ -\beta_\kappa & \mathbf{0} \end{pmatrix} \quad (40)$$

by a real orthogonal transformation $\mathbb{O}_\kappa$, where

$$\alpha_\kappa = \begin{pmatrix} \alpha_{\kappa 1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \alpha_{\kappa d_\kappa} \end{pmatrix} \text{ and } \beta_\kappa = \begin{pmatrix} \beta_{\kappa 1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \beta_{\kappa d_\kappa} \end{pmatrix} \quad (41)$$

as shown in Appendix B.

The same transformation thus acts on $\mathbb{B}_\kappa$ to produce

$$\mathbb{O}_\kappa^t \mathbb{B}_\kappa \mathbb{O}_\kappa = \mathbb{O}_\kappa^t \{\mathbb{N}_\kappa + i\mathbb{M}_\kappa\} \mathbb{O}_\kappa = \begin{pmatrix} \mathbf{0} & \alpha_\kappa + i\beta_\kappa \\ -(\alpha_\kappa + i\beta_\kappa) & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & e^{i\theta_\kappa} \\ -e^{i\theta_\kappa} & \mathbf{0} \end{pmatrix} \quad (42)$$

and thus

$$\mathbb{O}_\kappa^t \mathbb{C}_\kappa \mathbb{O}_\kappa = \eta_\kappa^{\frac{1}{2}} \begin{pmatrix} \mathbf{0} & e^{i\theta_\kappa} \\ -e^{i\theta_\kappa} & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \eta_\kappa^{\frac{1}{2}} e^{i\theta_\kappa} \\ -\eta_\kappa^{\frac{1}{2}} e^{i\theta_\kappa} & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \zeta_\kappa \\ -\zeta_\kappa & \mathbf{0} \end{pmatrix} \quad (43)$$

(which is valid even when $\eta_\kappa = 0$) [Note that $\theta_\kappa \equiv (\theta_{\kappa 1}, \dots, \theta_{\kappa d_\kappa})$ and $\theta_{\kappa i} \neq \theta_{\kappa j}$ in general] where

$$\zeta_\kappa = \begin{pmatrix} \zeta_{\kappa 1} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \zeta_{\kappa d_\kappa} \end{pmatrix}. \quad (44)$$

This allows us to define

$$\mathbb{O} = \begin{pmatrix} \mathbb{O}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbb{O}_p \end{pmatrix} \quad (45)$$

and see that

$$\mathbb{O}^t \mathbf{c}_U \mathbb{O} = \begin{pmatrix} \mathbf{0} & \zeta_1 & \mathbf{0} & \mathbf{0} \\ -\zeta_1 & \mathbf{0} & & \\ \mathbf{0} & & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \zeta_p \\ & & -\zeta_p & \mathbf{0} \end{pmatrix} \quad (46)$$

by rearranging the columns of $\mathbb{O}$ to produce $\tilde{\mathbb{O}}$ we obtain

$$\tilde{\mathbb{O}}^t \mathbf{c}_U \tilde{\mathbb{O}} = \begin{pmatrix} \mathbf{0} & \zeta \\ -\zeta & \mathbf{0} \end{pmatrix} \quad (47)$$

and hence using $\mathbf{c}_U = \mathbb{U}^\dagger \mathbf{c} \mathbb{U}^*$ we conclude that

$$\mathbb{T} \mathbf{c} \mathbb{T}^t = \tilde{\mathbb{O}}^t \mathbb{U}^\dagger \mathbf{c} \mathbb{U}^* \tilde{\mathbb{O}} = \begin{pmatrix} \mathbf{0} & \zeta \\ -\zeta & \mathbf{0} \end{pmatrix} \quad (48)$$

and define $\mathbb{T}$ as the unitary matrix

$$\mathbb{T} = \tilde{\mathbb{O}}^t \mathbb{U}^\dagger \quad (49)$$

where $\mathbb{T} \mathbb{T}^\dagger = \mathbb{T}^\dagger \mathbb{T} = \mathbb{I}_r$. ■

We now derive an important corollary that will be very useful in the physical interpretation of the density operators and expectation value expressions of AGP states:

Corollary 2 *If $\mathbf{c}$ is a $2s \times 2s$ complex antisymmetric matrix then there exists a unitary matrix $\mathbb{W}$ such that*

$$\mathbb{W} \mathbf{c} \mathbb{W}^t = \begin{pmatrix} \mathbf{0} & \mathbf{n}^{\frac{1}{2}} \\ -\mathbf{n}^{\frac{1}{2}} & \mathbf{0} \end{pmatrix} \quad (50)$$

where

$$\mathbf{n}^{\frac{1}{2}} = \begin{pmatrix} n_1^{\frac{1}{2}} & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & n_s^{\frac{1}{2}} \end{pmatrix}; \quad n_j^{\frac{1}{2}} \in \mathbb{R}, \quad 1 \leq j \leq s \quad (51)$$

and $\{n_j\}$ are the eigenvalues of $\mathbf{c} \mathbf{c}^\dagger$.

Proof. Using eq. (46) we can define $\mathbb{W}$ as

$$\mathbb{W} = \begin{pmatrix} e^{-i\frac{\theta_1}{2}} & \mathbf{0} & \cdots & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & e^{-i\frac{\theta_1}{2}} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & \mathbf{0} & \ddots & \cdots & \vdots & \vdots \\ \vdots & \vdots & \mathbf{0} & \ddots & \mathbf{0} & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & e^{-i\frac{\theta_p}{2}} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \cdots & \cdots & \mathbf{0} & e^{-i\frac{\theta_p}{2}} \end{pmatrix} \mathbb{T} \quad (52)$$

where

$$\mathbb{W}^\dagger \mathbb{W} = \mathbb{W} \mathbb{W}^\dagger = \mathbb{I}_{2s} \quad (53)$$

and

$$\mathbf{c} = \mathbb{W}^\dagger \begin{pmatrix} \mathbf{0} & \mathbf{n}^{\frac{1}{2}} \\ -\mathbf{n}^{\frac{1}{2}} & \mathbf{0} \end{pmatrix} \mathbb{W}^* = \mathbb{T}^\dagger \begin{pmatrix} \mathbf{0} & \zeta \\ -\zeta & \mathbf{0} \end{pmatrix} \mathbb{T}^* \quad (54)$$

■

Eq. (54) shows that the canonical NGSOs $\{|\psi_k\rangle\}$ are represented by the columns of the matrix $\mathbb{W}^\dagger$ in the basis $\{|\varphi_k\rangle\}$ giving

$$\begin{aligned} W(|\psi\rangle) &= (|\varphi\rangle) \\ W^\dagger(|\varphi\rangle) &= (|\varphi\rangle) \mathbb{W}^\dagger = (|\psi\rangle) \\ W^\dagger(|\varphi\rangle) &= (|\varphi\rangle) \begin{pmatrix} \mathbf{w}_1^\dagger & \cdots & \mathbf{w}_{2s}^\dagger \end{pmatrix} \end{aligned} \quad (55)$$

where

$$\begin{aligned} |g\rangle &= \frac{1}{2} \sum_{1 \leq k_1, k_2 \leq r} c_{k_1 k_2} |\varphi_{k_1} \varphi_{k_2}\rangle = \frac{1}{2} \sum_{1 \leq k_1, k_2 \leq r} \left(\mathbb{W}^\dagger \begin{pmatrix} \mathbf{0} & \mathbf{n}^{\frac{1}{2}} \\ -\mathbf{n}^{\frac{1}{2}} & \mathbf{0} \end{pmatrix} \mathbb{W}^* \right)_{k_1 k_2} |\varphi_{k_1} \varphi_{k_2}\rangle \\ &= \sum_{1 \leq k_1 \leq s} n_{k_1}^{\frac{1}{2}} |W^\dagger \varphi_{k_1} W^\dagger \varphi_{k_1+s}\rangle = \sum_{1 \leq k_1 \leq s} n_{k_1}^{\frac{1}{2}} |\psi_{k_1} \psi_{k_1+s}\rangle \end{aligned} \quad (56)$$

One can obtain a non unique canonical expansion by setting

$$\begin{aligned} |g\rangle &= \sum_{1 \leq k_1 \leq s} n_{k_1}^{\frac{1}{2}} \left| e^{i\frac{\theta_{k_1}}{2}} T^\dagger \varphi_{k_1} e^{i\frac{\theta_{k_1}}{2}} T^\dagger \varphi_{k_1+s} \right\rangle = \sum_{1 \leq k_1 \leq s} n_{k_1}^{\frac{1}{2}} e^{i\theta_{k_1}} |T^\dagger \varphi_{k_1} T^\dagger \varphi_{k_1+s}\rangle \\ &= \sum_{1 \leq k_1 \leq s} n_{k_1}^{\frac{1}{2}} e^{i\theta_{k_1}} |\chi_{k_1} \chi_{k_1+s}\rangle \end{aligned} \quad (57)$$

so that

$$\begin{aligned} T(|\chi\rangle) &= (|\varphi\rangle) \\ T^\dagger(|\chi\rangle) &= (|\varphi\rangle) \mathbb{T}^\dagger = (|\chi\rangle) \end{aligned} \quad (58)$$

The non uniqueness of the complex canonical coefficients is easily seen by noting that

$$|g\rangle = \sum_{1 \leq k_1 \leq s} n_{k_1}^{\frac{1}{2}} e^{i\theta_{k_1}} e^{-i\nu_{k_1}} \left| e^{i\frac{\nu_{k_1}}{2}} \chi_{k_1} e^{i\frac{\nu_{k_1}}{2}} \chi_{k_1+s} \right\rangle = \sum_{1 \leq k_1 \leq s} n_{k_1}^{\frac{1}{2}} e^{i(\theta_{k_1} - \nu_{k_1})} |\tilde{\chi}_{k_1} \tilde{\chi}_{k_1+s}\rangle \quad (59)$$

where

$$|\tilde{\chi}_{k_1}\rangle = \left| e^{i\frac{\nu_{k_1}}{2}} \chi_{k_1} \right\rangle \quad (60)$$

We now prove the often stated fact that the eigenvalues of $\mathbb{D}_\varphi^1(g)$ are at least doubly degenerate

Corollary 3 *The matrix $\mathbf{c}\mathbf{c}^\dagger$ has eigenvalues that are degenerate of degree $2m$; $1 \leq m \leq s$.*

Proof. Eq. (29) shows that if the eigenvalue n_j is non degenerate then $c_{U_{jk}} = 0$; $1 \leq k \leq r$. Eq. (34) shows in particular that for such non degenerate eigenvalues $|c_{U_{jj}}|^2 = n_j = 0$, thus all non degenerate eigenvalues must be equal to 0.

If an eigenvalue is doubly degenerate then the associated unitary antisymmetric block $\mathbb{B}$ is 2×2 and it is unitary and antisymmetric, requirements that it can easily be satisfied by non zero $\mathbb{B}$'s. This is true for all evenly degenerate eigenvalues.

If an eigenvalue is triply degenerate then the associated unitary antisymmetric block $\mathbb{B}$ is 3×3 . As it is antisymmetric one or all three of its eigenvalues must be 0 (as by Appendix C the eigenvalues must be +/- pairs or zero) which is a contradiction as the matrix $\mathbb{B}$ is also unitary so cannot have zero eigenvalues. Thus the block cannot be 3×3 , the same argument holds for any odd dimensioned block hence only even dimensional blocks can exist. ■

IV. NUMERICAL CONSTRUCTION OF CANONICAL GSO

The first step is to construct the FORDM $\mathbb{D}_\varphi^1(g) = \frac{1}{2}\mathbf{c}\mathbf{c}^\dagger$ then diagonalize it by a unitary transformation $\mathbb{Y}$. One then transforms $\mathbf{c}$ by $\mathbb{Y}^\dagger \mathbf{c} \mathbb{Y}^*$ into block diagonal form with blocks $\{\mathbb{C}_\kappa\}$ each associated with a degenerate eigenvalue η_κ of $\mathbb{D}_\varphi^1(g)$. A unitary antisymmetric matrix $\mathbb{B}_\kappa$ is then produced from each block $\mathbb{C}_\kappa$ associated with a non zero eigenvalue by

$$\mathbb{B}_\kappa = \frac{1}{\sqrt{\eta_\kappa}} \mathbf{c}_\kappa \quad (61)$$

The next step is to produce a unitary transformation that diagonalizes $\mathbb{B}_\kappa$. One procedure to achieve this would be simultaneously diagonalize the self adjoint real and imaginary parts

of $\mathbb{B}$

$$\begin{aligned}\mathbb{B}_R &= \text{Re } \mathbb{B} = \frac{1}{2} \{ \mathbb{B} + \mathbb{B}^\dagger \} \\ \mathbb{B}_I &= \text{Im } \mathbb{B} = -\frac{i}{2} \{ \mathbb{B} - \mathbb{B}^\dagger \}\end{aligned}\tag{62}$$

which are both antisymmetric

$$\mathbb{B}_R = -\mathbb{B}_R^t \text{ and } \mathbb{B}_I = -\mathbb{B}_I^t\tag{63}$$

and commute with each other

$$[\mathbb{B}_R, \mathbb{B}_I] = 0\tag{64}$$

where we have dropped the subscript κ . As neither matrix is positive definite the standard way to do this is to first diagonalize one, then diagonalize the other in the eigen basis of the first. However if the dimension of the nonzero blocks $\{\mathbb{B}_R, \mathbb{B}_I\}$ are greater than 2 e.g. 4, 6, $\dots$, etc., there are more than 1 positive and 1 negative eigenvalues (there are always equal numbers of positive and negative eigenvalues as $\{\mathbb{B}_R, \mathbb{B}_I\}$ are antisymmetric), and numerical diagonalization routines will order these positive and negative sets in some predetermined sequence. We know from eqs. (40) and (41) and Appendix B that in order to obtain a single unitary transformation that produces a single orthogonal transformation $\mathbb{O}$ that simultaneously brings $\{\mathbb{B}_R, \mathbb{B}_I\}$ to a canonical form the eigenvalues $\{\mp i\alpha_j, \mp i\beta_j; 1 \leq j \leq d\}$ of $\mathbb{B}_R$ and $\mathbb{B}_I$ must correspond in a very precise manner, so one would need to find and apply some principle of correspondence and a procedure that to accomplish this.

One can avoid this problem by directly diagonalizing the antisymmetric normal, (therefore diagonalizable), unitary matrix $\mathbb{B}$ with a unitary matrix $\mathbb{V}$ to obtain

$$\mathbb{B} = (\mathbf{v}_1 \cdots \mathbf{v}_d \mathbf{v}_1^* \cdots \mathbf{v}_d^*) \begin{pmatrix} -ie^{i\theta_1} & 0 & 0 & 0 & \cdots & 0 \\ 0 & \ddots & 0 & 0 & \cdots & 0 \\ 0 & \cdots & -ie^{i\theta_d} & 0 & \cdots & 0 \\ 0 & \cdots & 0 & ie^{i\theta_{d+1}} & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \ddots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & ie^{i\theta_{2d}} \end{pmatrix} \begin{pmatrix} \mathbf{v}_1 \\ \vdots \\ \mathbf{v}_d \\ \mathbf{v}_1^* \\ \vdots \\ \mathbf{v}_d^* \end{pmatrix}\tag{65}$$

The disadvantage in choosing this route is that one must use a general complex matrix diagonalization routine instead of complex self adjoint ones, which of course incurs more computer costs. A further numerical problem now becomes apparent, although one can prove

that if $\mathbf{v}$ is an eigenvector of $\mathbb{B}$ with eigenvalue $-ie^{i\theta}$ then $\mathbf{v}^*$ is an eigenvector with eigenvalue $ie^{i\theta}$ (see Appendix C), the eigenvectors, even in absence of degeneracies, are only unique up to an overall phase factor and thus they do not have to be produced in such complex conjugate pairs in standard general complex matrix diagonalization routines. Furthermore if there are degeneracies the eigenvectors corresponding to the degenerate eigenvalue do not have to be orthogonal, though it is still true that for normal matrices eigenvectors corresponding to different eigenvalues are, like the hermitian case, orthogonal. The way to overcome this non orthogonality and pairing problem is to first Löwdin symmetric orthogonalize [?] the degenerate eigenvectors corresponding to the eigenvalue that has a negative real part and then *define* the eigenvectors associated with the negative of this eigenvalue as the complex conjugates of these eigenvectors.

Once a paired unitary matrix $\mathbb{V}$ has been produced one can form a real orthogonal $2d \times 2d$ matrix $\mathbb{O}$ by defining its columns by

$$\begin{aligned}\mathbf{o}_j &= \frac{1}{\sqrt{2}} (\mathbf{v}_j + \mathbf{v}_j^*) = \sqrt{2} \operatorname{Re} \mathbf{v}_j \\ \mathbf{o}_{j+d} &= \frac{-i}{\sqrt{2}} (\mathbf{v}_j - \mathbf{v}_j^*) = \sqrt{2} \operatorname{Im} \mathbf{v}_j\end{aligned}\tag{66}$$

which brings the complex antisymmetric unitary matrix $\mathbb{B}$ to a canonical form, allowing one to express $\mathbb{B}$ as

$$\mathbb{B} = (\mathbf{o}_1 \cdots \mathbf{o}_d \mathbf{o}_{d+1} \cdots \mathbf{o}_{2d}) \begin{pmatrix} 0 & 0 & 0 & ie^{i\theta_1} & \cdots & 0 \\ 0 & \ddots & 0 & 0 & \ddots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & ie^{i\theta_d} \\ -ie^{i\theta_{d+1}} & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \ddots & 0 & 0 & \ddots & 0 \\ 0 & \cdots & -ie^{i\theta_{2d}} & 0 & \cdots & 0 \end{pmatrix} \begin{pmatrix} \mathbf{o}_1 \\ \vdots \\ \mathbf{o}_d \\ \mathbf{o}_{d+1} \\ \vdots \\ \mathbf{o}_{2d} \end{pmatrix}\tag{67}$$

This allows us to construct a unitary matrix $\tilde{\mathbb{O}}$ that brings $\mathbb{B}$ to a unique canonical form by

defining

$$(\tilde{\mathbf{o}}_1 \cdots \tilde{\mathbf{o}}_d \tilde{\mathbf{o}}_{d+1} \cdots \tilde{\mathbf{o}}_{2d}) = (\mathbf{o}_1 \cdots \mathbf{o}_d \mathbf{o}_{d+1} \cdots \mathbf{o}_{2d}) \begin{pmatrix} e^{\frac{i\theta_1}{2}} & 0 & 0 & 0 & \cdots & 0 \\ 0 & \ddots & 0 & 0 & \cdots & 0 \\ 0 & \cdots & e^{\frac{i\theta_d}{2}} & 0 & \cdots & 0 \\ 0 & \cdots & 0 & e^{\frac{i\theta_{d+1}}{2}} & \cdots & 0 \\ 0 & \cdots & 0 & 0 & \ddots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & e^{\frac{i\theta_{2d}}{2}} \end{pmatrix} \quad (68)$$

which leads to

$$\mathbb{B} = (\tilde{\mathbf{o}}_1 \cdots \tilde{\mathbf{o}}_d \tilde{\mathbf{o}}_{d+1} \cdots \tilde{\mathbf{o}}_{2d}) \begin{pmatrix} 0 & 0 & 0 & 1 & \cdots & 0 \\ 0 & \ddots & 0 & 0 & \ddots & 0 \\ 0 & \cdots & 0 & 0 & \cdots & 1 \\ -1 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \ddots & 0 & 0 & \ddots & 0 \\ 0 & \cdots & -1 & 0 & \cdots & 0 \end{pmatrix} \begin{pmatrix} \tilde{\mathbf{o}}_1 \\ \vdots \\ \tilde{\mathbf{o}}_d \\ \tilde{\mathbf{o}}_{d+1} \\ \vdots \\ \tilde{\mathbf{o}}_{2d} \end{pmatrix} \quad (69)$$

and hence

$$\mathbf{c}_\kappa = (\tilde{\mathbf{o}}_1 \cdots \tilde{\mathbf{o}}_{d_\kappa} \tilde{\mathbf{o}}_{d_\kappa+1} \cdots \tilde{\mathbf{o}}_{2d_\kappa}) \begin{pmatrix} \mathbf{0} & \eta_\kappa^{\frac{1}{2}} \mathbb{I}_{d_\kappa} \\ -\eta_\kappa^{\frac{1}{2}} \mathbb{I}_{d_\kappa} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \tilde{\mathbf{o}}_1 \\ \vdots \\ \tilde{\mathbf{o}}_{d_\kappa} \\ \tilde{\mathbf{o}}_{d_\kappa+1} \\ \vdots \\ \tilde{\mathbf{o}}_{2d_\kappa} \end{pmatrix} \quad (70)$$

and the full blocked diagonal $\mathbf{c}$ matrix

$$\begin{aligned} \mathbb{Y}^\dagger \mathbf{c} \mathbb{Y}^* &= \begin{pmatrix} \mathbf{c}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{c}_p \end{pmatrix} \\ &= \begin{pmatrix} \tilde{\mathbb{O}}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \tilde{\mathbb{O}}_p \end{pmatrix} \begin{pmatrix} \eta_1^{\frac{1}{2}} \mathbb{J}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \eta_p^{\frac{1}{2}} \mathbb{J}_p \end{pmatrix} \begin{pmatrix} \tilde{\mathbb{O}}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \tilde{\mathbb{O}}_p \end{pmatrix}^t \end{aligned} \quad (71)$$

where

$$\mathbb{J}_\kappa = \begin{pmatrix} \mathbf{0} & \mathbb{I}_{d_\kappa} \\ -\mathbb{I}_{d_\kappa} & \mathbf{0} \end{pmatrix} \quad (72)$$

This leads to

$$\mathbf{c} = \mathbb{Y} \tilde{\mathbb{O}} \begin{pmatrix} \eta_1^{\frac{1}{2}} \mathbb{J}_1 & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \eta_p^{\frac{1}{2}} \mathbb{J}_p \end{pmatrix} \tilde{\mathbb{O}}^t \mathbb{Y}^* \quad (73)$$

and hence the unitary matrix $\mathbb{W}^\dagger$ whose columns are the NGSOs by

$$\mathbb{W}^\dagger = \mathbb{Y} \tilde{\mathbb{O}} \quad (74)$$

V. DISCUSSION

We have shown in the preceding that one can form unique canonical coefficients for a geminal but one must be very careful in the numerical implementation of the constructions contained in Zuminos' Theorem, i.e. one must constrain the computed quantities to correspond to the desired mathematical ones. These unique coefficients have physical significance as they determine the eigenvalues i.e. occupation numbers of the FORDO of the $2N$ -particle AGP state *and* the full Configurational Interaction (CI) expansion. A rather remarkable consequence of this is that the energy of the $2N$ -particle AGP state can solely be expressed in terms of the one particle canonical orbitals and the occupation numbers of the geminal.

VI. ACKNOWLEDGEMENTS

We would like thank the support of the Petroleum Research Foundation through a grant PRF 34859-AC6.

APPENDIX A: TENSOR PRODUCTS

In order to use familiar terms we will use the coordinate representation of one particle states $\{|\varphi\rangle\}$ in terms of wavefunctions $\varphi(\mathbf{r})$ to describe the various tensor products. The coordinate representation is based on the generalized eigenvectors $\{|\mathbf{r}\rangle\}$ of the position operator that produce the resolution of the identity

$$I = \int |\mathbf{r}\rangle \langle \mathbf{r}| d\mathbf{r} \quad (\text{A1})$$

by

$$|\varphi\rangle = \int |\mathbf{r}\rangle \langle \mathbf{r}|\varphi\rangle d\mathbf{r} = \int |\mathbf{r}\rangle \varphi(\mathbf{r}) d\mathbf{r} \quad (\text{A2})$$

The tensor product $\varphi_1 \otimes \varphi_2(\mathbf{r}_1, \mathbf{r}_2)$ of two one particle state wavefunctions $\{\varphi_1(\mathbf{r}), \varphi_2(\mathbf{r})\}$ is the pointwise product

$$\varphi_1 \otimes \varphi_2(\mathbf{r}_1, \mathbf{r}_2) = \varphi_1(\mathbf{r}_1) \otimes \varphi_2(\mathbf{r}_2) = \varphi_1(\mathbf{r}_1) \varphi_2(\mathbf{r}_2) \quad (\text{A3})$$

this product has no particle symmetry but is the easiest to perform computations such as contractions in. If φ_1 and φ_2 are normalized then so is $\varphi_1 \otimes \varphi_2$. The two particle state wavefunction that is the antisymmetric tensor product $\varphi_1 \wedge \varphi_2(\mathbf{r}_1, \mathbf{r}_2)$ of two one particle state wavefunctions $\{\varphi_1(\mathbf{r}), \varphi_2(\mathbf{r})\}$ is given by

$$\varphi_1 \wedge \varphi_2(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} \{\varphi_1(\mathbf{r}_1) \otimes \varphi_2(\mathbf{r}_2) - \varphi_2(\mathbf{r}_1) \otimes \varphi_1(\mathbf{r}_2)\} \quad (\text{A4})$$

where the factor $\frac{1}{\sqrt{2}}$ has been introduced to maintain normalization and the RHS of eq. (A4) is recognized as a normalized Slater determinant of the two one particle wavefunctions $\{\varphi_1(\mathbf{r}), \varphi_2(\mathbf{r})\}$ i.e.

$$\varphi_1 \wedge \varphi_2(\mathbf{r}_1, \mathbf{r}_2) = \frac{1}{\sqrt{2}} \begin{vmatrix} \varphi_1(\mathbf{r}_1) & \varphi_1(\mathbf{r}_2) \\ \varphi_2(\mathbf{r}_1) & \varphi_2(\mathbf{r}_2) \end{vmatrix} \quad (\text{A5})$$

APPENDIX B: CANONICAL FORM OF A REAL ANTISYMMETRIC MATRIX

The commuting *real* antisymmetric matrices $\mathbb{N}$ and $\mathbb{M}$ define hermitian matrices

$$\begin{aligned} \mathbb{H}_N &= i\mathbb{N} \\ \mathbb{H}_M &= i\mathbb{M} \end{aligned} \quad (\text{B1})$$

which can be seen from

$$\begin{aligned} \mathbb{H}_N^\dagger &= (i\mathbb{N})^{t*} = (i\mathbb{N}^t)^* = (-i\mathbb{N})^* = i\mathbb{N} = \mathbb{H}_N \\ \mathbb{H}_M^\dagger &= (i\mathbb{M})^{t*} = (i\mathbb{M}^t)^* = (-i\mathbb{M})^* = i\mathbb{M} = \mathbb{H}_M \end{aligned} \quad (\text{B2})$$

The hermitian matrices $\mathbb{H}_N$ and $\mathbb{H}_M$ commute and thus can be simultaneously diagonalized into real diagonal forms by the same unitary transformation and hence we show in the following that $\mathbb{N}$ and $\mathbb{M}$ can simultaneously be brought to a canonical antisymmetric form by using the same real orthogonal transformation that is constructed according to the following:

Consider the real antisymmetric matrix $\mathbb{A} = \mathbb{A}^*$ it can be converted into a hermitian purely imaginary matrix, $\mathbb{H}$, with the properties

$$\begin{aligned}\mathbb{H} &= i\mathbb{A} \\ \mathbb{H}^\dagger &= -i\mathbb{A}^t = i\mathbb{A} = \mathbb{H} \\ \mathbb{H}^* &= -i\mathbb{A} = i\mathbb{A}^t = \mathbb{H}^t\end{aligned}\tag{B3}$$

and

$$\mathbb{H} = i\mathbb{A} = \sum_{1 \leq j \leq s} \alpha_j \{ |\mathbf{v}_j\rangle \langle \mathbf{v}_j| - |\mathbf{v}_j^*\rangle \langle \mathbf{v}_j^*| \}; \quad \alpha_j \in \mathbb{R}\tag{B4}$$

where $\{ |\mathbf{v}_j\rangle, |\mathbf{v}_j^*\rangle; 1 \leq j \leq r \}$ is a complete orthonormal basis. Eq. (B4) is seen by observing that

$$\mathbb{H} |\mathbf{v}\rangle = \alpha |\mathbf{v}\rangle \Rightarrow \mathbb{H}^* |\mathbf{v}^*\rangle = \alpha |\mathbf{v}^*\rangle \Rightarrow \mathbb{H}^t |\mathbf{v}^*\rangle = \alpha |\mathbf{v}^*\rangle \Rightarrow \mathbb{H} |\mathbf{v}^*\rangle = -\alpha |\mathbf{v}^*\rangle\tag{B5}$$

It is important to note that the orthonormal basis eigenvectors of $\mathbb{H}$ are only defined up to an overall phase factors $e^{i\varphi_j}$ which *do not* preserve the complex conjugate relationship between eigenvectors associated with paired eigenvalues of opposite sign, however one can always find complex conjugate eigenvector pairs. These different bases are related to each other by

$$(|\mathbf{v}'_1\rangle \cdots |\mathbf{v}'_{2s}\rangle) = (|\mathbf{v}_1\rangle \cdots |\mathbf{v}_{2s}\rangle) \begin{pmatrix} e^{i\varphi_1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \ddots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & e^{i\varphi_{2s}} \end{pmatrix}\tag{B6}$$

Using a basis with complex conjugate symmetry we obtain

$$\mathbb{A} = - \sum_{1 \leq j \leq s} i\alpha_j \{ |\mathbf{v}_j\rangle \langle \mathbf{v}_j| - |\mathbf{v}_j^*\rangle \langle \mathbf{v}_j^*| \}; \quad \alpha_i \in \mathbb{R}\tag{B7}$$

We can then define a *real* conb $\{ |\mathbf{s}_j\rangle; 1 \leq j \leq \infty \}$ i.e. $|\mathbf{s}_j\rangle = |\mathbf{s}_j^*\rangle$ as

$$\begin{aligned}|\mathbf{s}_j\rangle &= \frac{1}{\sqrt{2}} (|\mathbf{v}_j\rangle + |\mathbf{v}_j^*\rangle) = \sqrt{2} \text{Re} |\mathbf{v}_j\rangle \\ |\mathbf{s}_{j+s}\rangle &= \frac{-i}{\sqrt{2}} ((|\mathbf{v}_j\rangle - |\mathbf{v}_j^*\rangle)) = \sqrt{2} \text{Im} |\mathbf{v}_j\rangle\end{aligned}\tag{B8}$$

so that

$$\begin{aligned}\mathbb{A} \left\{ \frac{1}{\sqrt{2}} (|\mathbf{v}_j\rangle + |\mathbf{v}_j^*\rangle) \right\} &= -i\alpha_j \frac{1}{\sqrt{2}} (|\mathbf{v}_j\rangle - |\mathbf{v}_j^*\rangle) \\ &= \alpha_j \left\{ \frac{-i}{\sqrt{2}} ((|\mathbf{v}_j\rangle - |\mathbf{v}_j^*\rangle)) \right\}\end{aligned}\tag{B9}$$

and

$$\begin{aligned}\mathbb{A} \left\{ \frac{-i}{\sqrt{2}} (| \mathbf{v}_j \rangle - | \mathbf{v}_j^* \rangle) \right\} &= \frac{-\alpha_j}{\sqrt{2}} | \mathbf{v}_j \rangle - \frac{\alpha_j}{\sqrt{2}} | \mathbf{v}_j^* \rangle \\ &= -\alpha_j \left\{ \frac{1}{\sqrt{2}} (| \mathbf{v}_j \rangle + | \mathbf{v}_j^* \rangle) \right\}\end{aligned}\quad (\text{B10})$$

thus

$$\mathbb{A} = \sum_{1 \leq j \leq s} \alpha_j \{ | \mathbf{s}_j \rangle \langle \mathbf{s}_{j+s} | - | \mathbf{s}_{j+s} \rangle \langle \mathbf{s}_j | \} \quad (\text{B11})$$

which displays the desired result. Using the bases $\{ | \mathbf{s}_j \rangle ; 1 \leq j \leq \infty \}$ we can produce the canonical matrix representation of $\mathbb{A}$

$$\mathbb{A} (| \mathbf{s} \rangle) = | \mathbf{s} \rangle \begin{pmatrix} \mathbb{O} & \alpha \\ -\alpha & \mathbb{O} \end{pmatrix} \quad (\text{B12})$$

where

$$\alpha = \begin{pmatrix} \alpha_1 & \mathbb{O} & \mathbb{O} \\ \mathbb{O} & \ddots & \mathbb{O} \\ \mathbb{O} & \mathbb{O} & \alpha_s \end{pmatrix} \quad (\text{B13})$$

and we have the expansion

$$\mathbb{A} = | \mathbf{s} \rangle \begin{pmatrix} \mathbb{O} & \alpha \\ -\alpha & \mathbb{O} \end{pmatrix} \langle \mathbf{s} | \quad (\text{B14})$$

The row vector of basis vectors, $(| \mathbf{s}_1 \rangle \cdots | \mathbf{s}_s \rangle | \mathbf{s}_{s+1} \rangle \cdots | \mathbf{s}_{s+s} \rangle)$, can be viewed as a real orthogonal matrix whose columns are the real basis vectors $\{ | \mathbf{s}_1 \rangle, \cdots | \mathbf{s}_s \rangle, | \mathbf{s}_{s+1} \rangle, \cdots | \mathbf{s}_{s+s} \rangle \}$.

The complex basis $\{ | \mathbf{v}_j \rangle, | \mathbf{v}_j^* \rangle ; 1 \leq j \leq r \}$ diagonalizes $\mathbb{A}$ to give

$$\mathbb{A} (| \mathbf{v} \rangle, | \mathbf{v}^* \rangle) = (| \mathbf{v} \rangle, | \mathbf{v}^* \rangle) \begin{pmatrix} -i\alpha & \mathbb{O} \\ \mathbb{O} & i\alpha \end{pmatrix} \quad (\text{B15})$$

and produces the expansion

$$A = (| \mathbf{v} \rangle, | \mathbf{v}^* \rangle) \begin{pmatrix} -i\alpha & \mathbb{O} \\ \mathbb{O} & i\alpha \end{pmatrix} \begin{pmatrix} \langle \mathbf{v} | \\ \langle \mathbf{v}^* | \end{pmatrix} \quad (\text{B16})$$

The row vector of basis vectors, $(| \mathbf{v}_1 \rangle \cdots | \mathbf{v}_s \rangle | \mathbf{v}_1^* \rangle \cdots | \mathbf{v}_s^* \rangle)$, can be viewed as an unitary matrix whose columns are the basis vectors $\{ | \mathbf{v}_1 \rangle \cdots | \mathbf{v}_s \rangle | \mathbf{v}_1^* \rangle \cdots | \mathbf{v}_s^* \rangle \}$. Note that the expansions of eqs. (B14) and (B16) both give $\mathbb{A}$ thus in eq. (B14) all the imaginary terms cancel.

APPENDIX C: COMPLEX CONJUGATE PAIRING

Let $\mathbb{A}$ be a complex antisymmetric matrix that is normal then

$$[\mathbb{A}, \mathbb{A}^\dagger] = 0 \quad (\text{C1})$$

and $\mathbb{A}$ and $\mathbb{A}^\dagger$ are simultaneously diagonalizable and the spectral resolution gives

$$\mathbb{A} = \sum_{1 \leq j \leq r} \alpha_j |\mathbf{v}_j\rangle \langle \mathbf{v}_j| \quad (\text{C2a})$$

$$\mathbb{A}^\dagger = \sum_{1 \leq j \leq r} \alpha_j^* |\mathbf{v}_j\rangle \langle \mathbf{v}_j| \quad (\text{C2b})$$

so there exist simultaneous eigenvectors $\{\mathbf{v}_j\}$ to $\mathbb{A}$ and $\mathbb{A}^\dagger$ corresponding to complex conjugate eigenvalues $\{\alpha, \alpha^*\}$ i.e.

$$\begin{aligned} \mathbb{A} |\mathbf{v}_j\rangle &= \alpha |\mathbf{v}_j\rangle \\ \mathbb{A}^\dagger |\mathbf{v}_j\rangle &= \alpha_j^* |\mathbf{v}_j\rangle \end{aligned} \quad (\text{C3a})$$

as $\mathbb{A}$ is antisymmetric we obtain from eq. (C2b) that

$$\mathbb{A}^* |\mathbf{v}_j\rangle = -\alpha_j^* |\mathbf{v}_j\rangle \quad (\text{C4})$$

which gives by complex conjugation that

$$\mathbb{A} |\mathbf{v}_j^*\rangle = -\alpha_j |\mathbf{v}_j^*\rangle \quad (\text{C5})$$